ISA Data Fluency

IMBA Session 7

"This is my favorite part about analytics: taking boring flat data and bringing it to life through visualization." –John Tukey

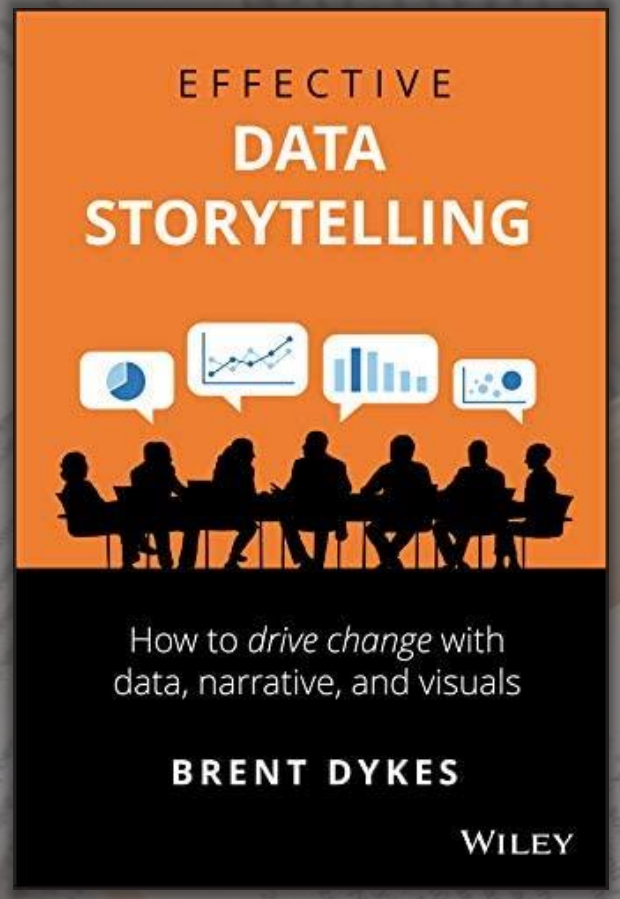
Professor Christina Stathopoulos, October 2026

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Session 5

Book of the Day

Effective Data Storytelling by Brent Dykes

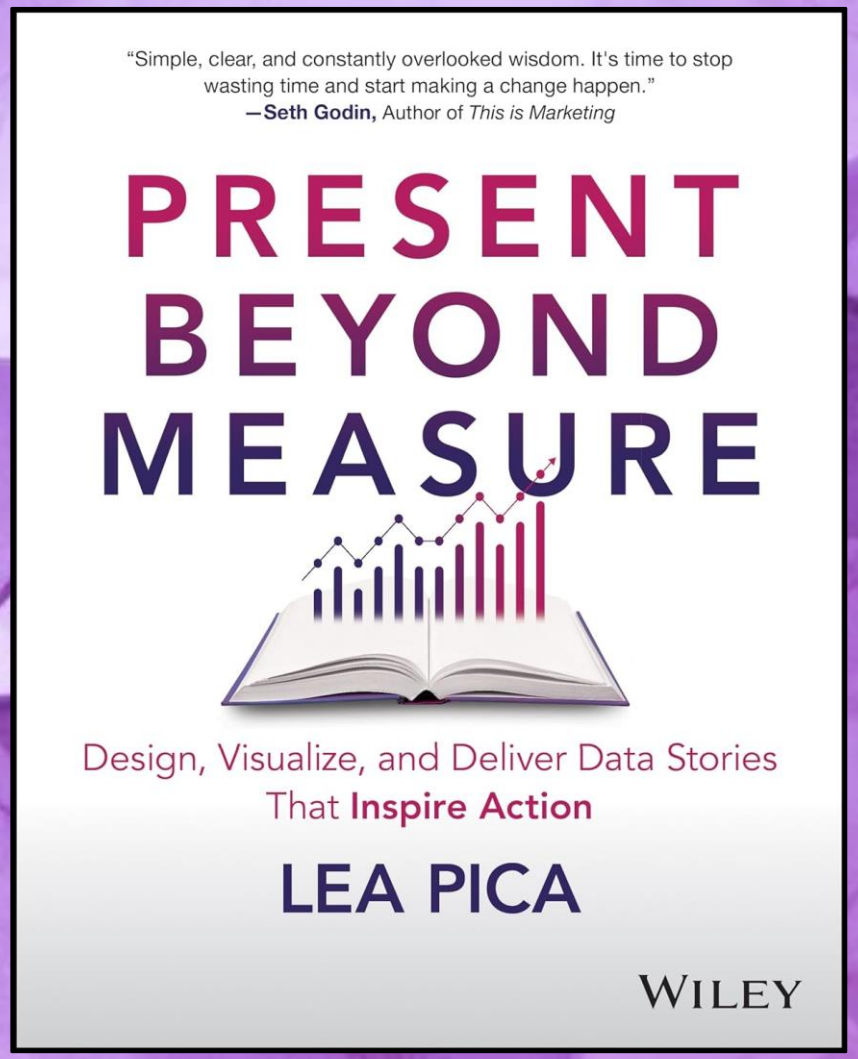


Session 6

Book of the Day

Present Beyond Measure by Lea Pica





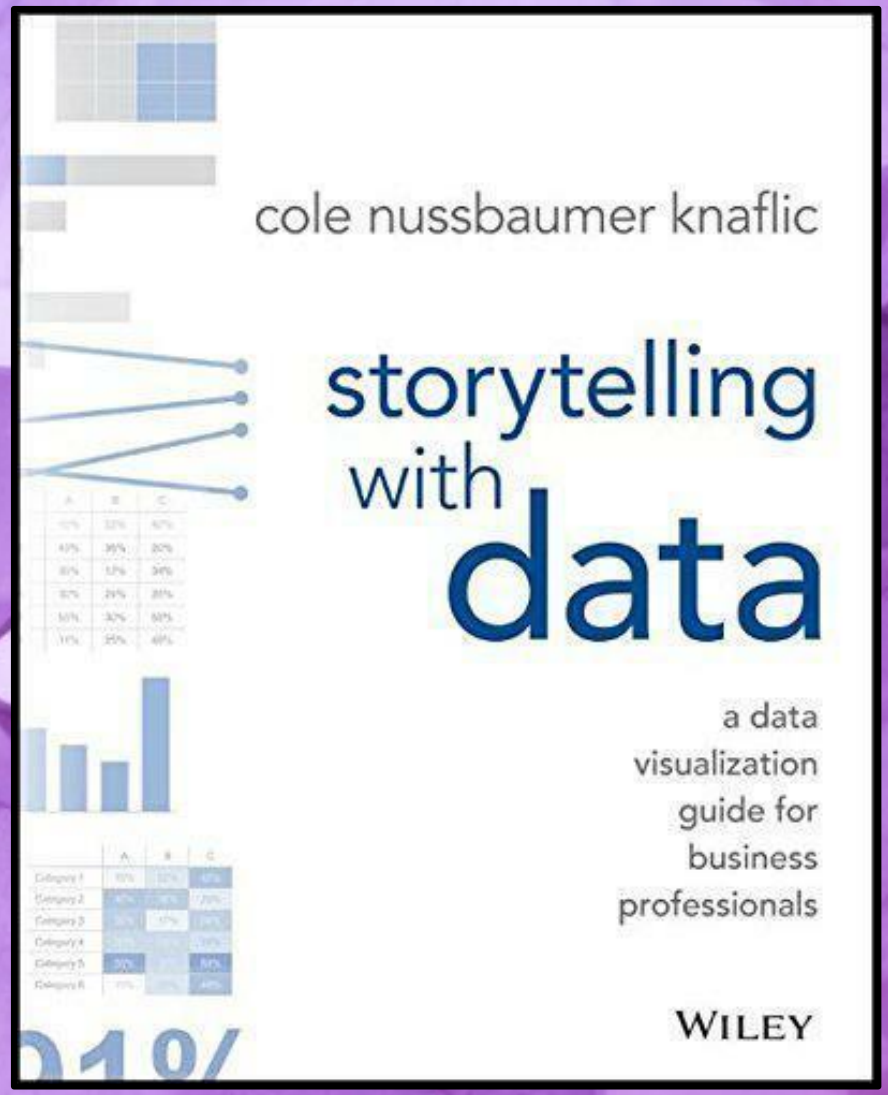
Book of the Day

Session 7

Storytelling with Data

by Cole Nussbaumer Knaflic





Session 7



Intro to [Data] Storytelling



Inspiration

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Intro to [Data] Storytelling

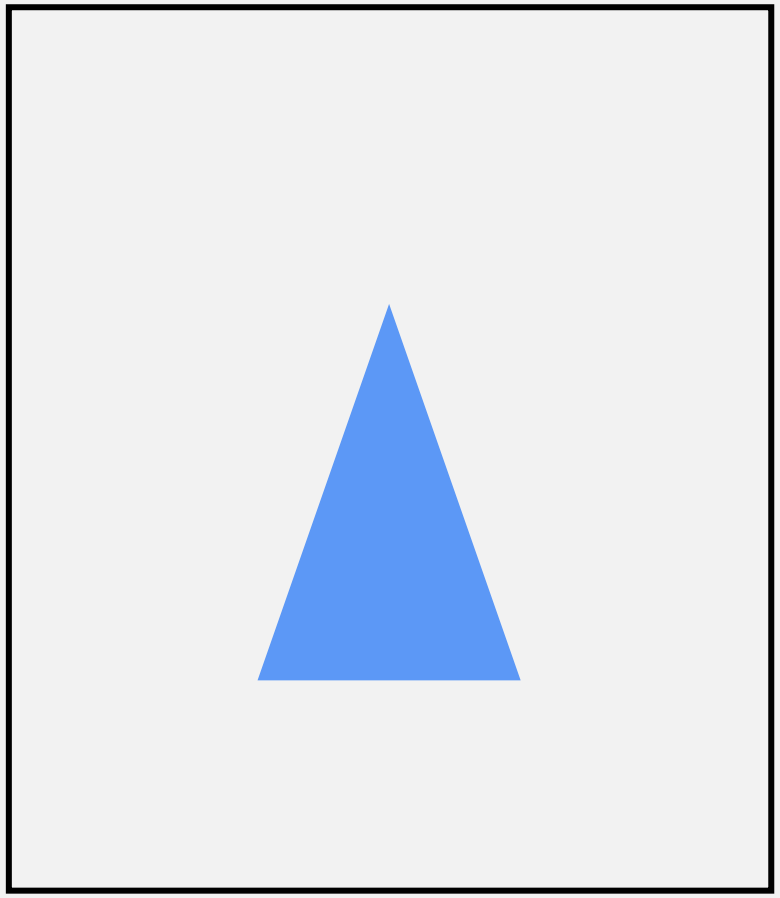
Bringing your data to life



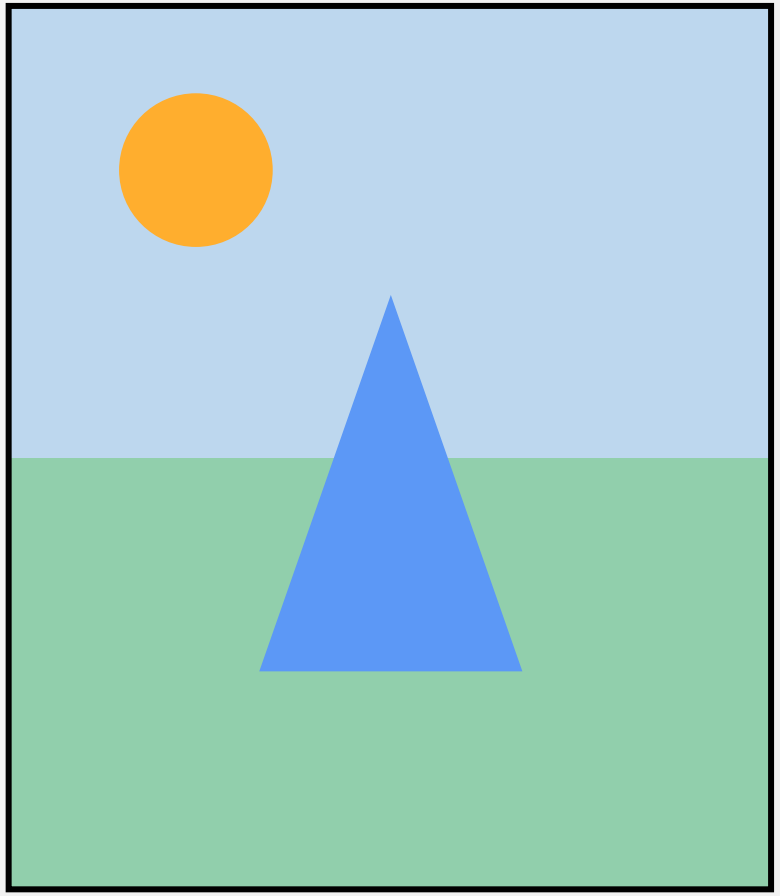
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Why do we need to add storytelling to data?





Data



+ Storytelling

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Telling a data story correctly can save lives



- The data tables did not reveal the cause

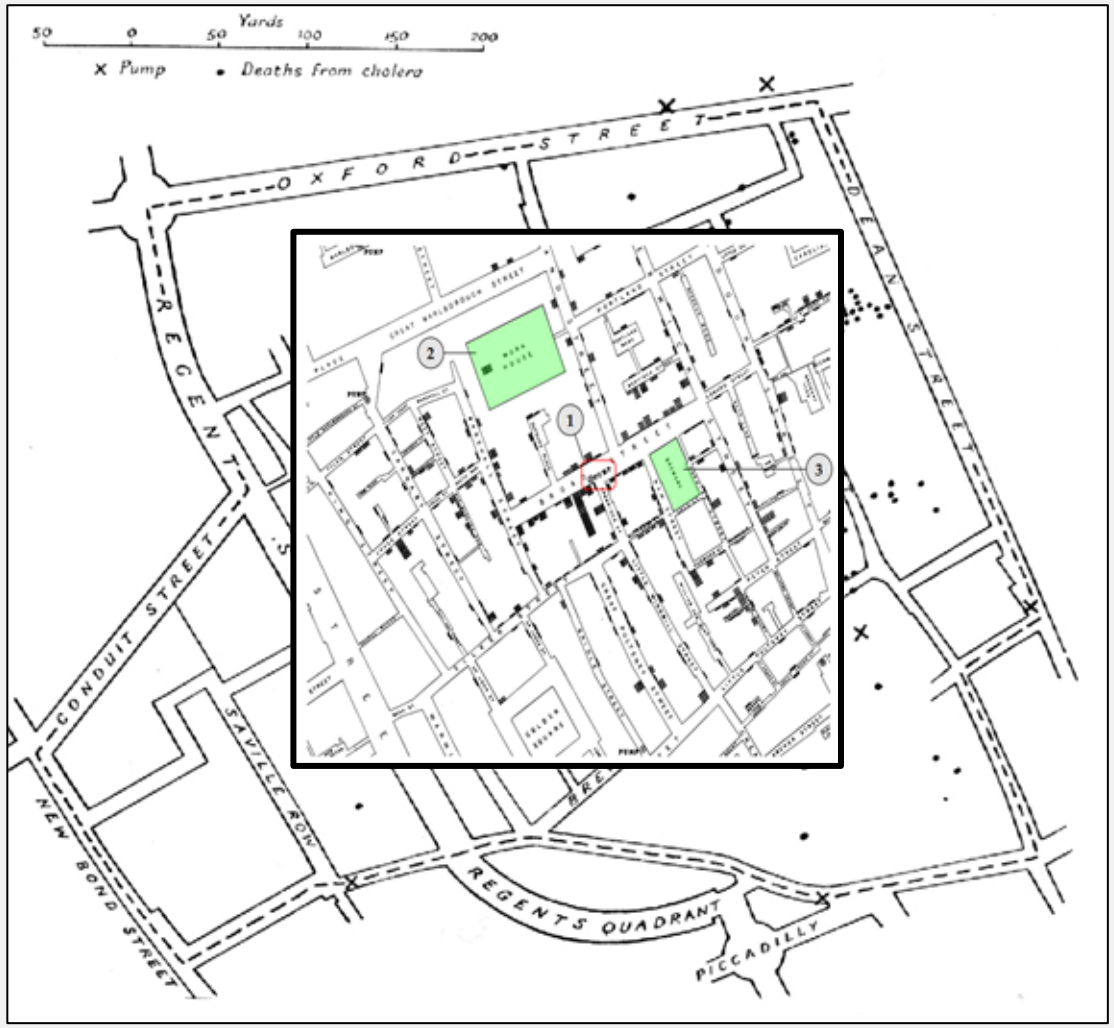


Dr. John Snow

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Telling a data story correctly can save lives



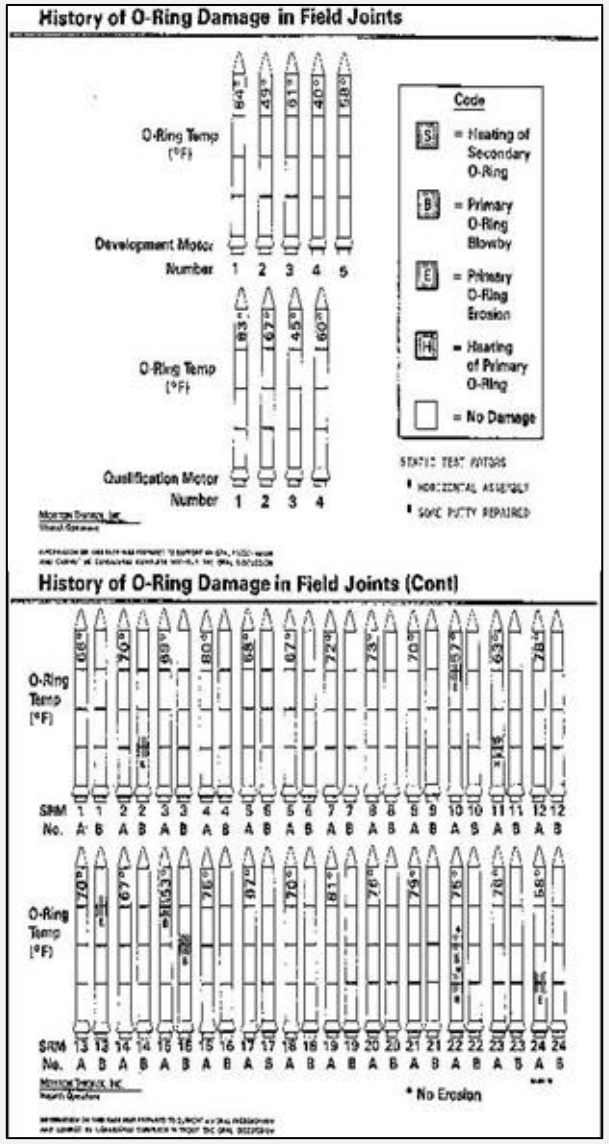
- The data tables did not reveal the cause.
- But the map of cases revealed a clear story.
- They removed the water pump handle and the outbreak ended.

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Telling a data story correctly can save lives

…or cost them



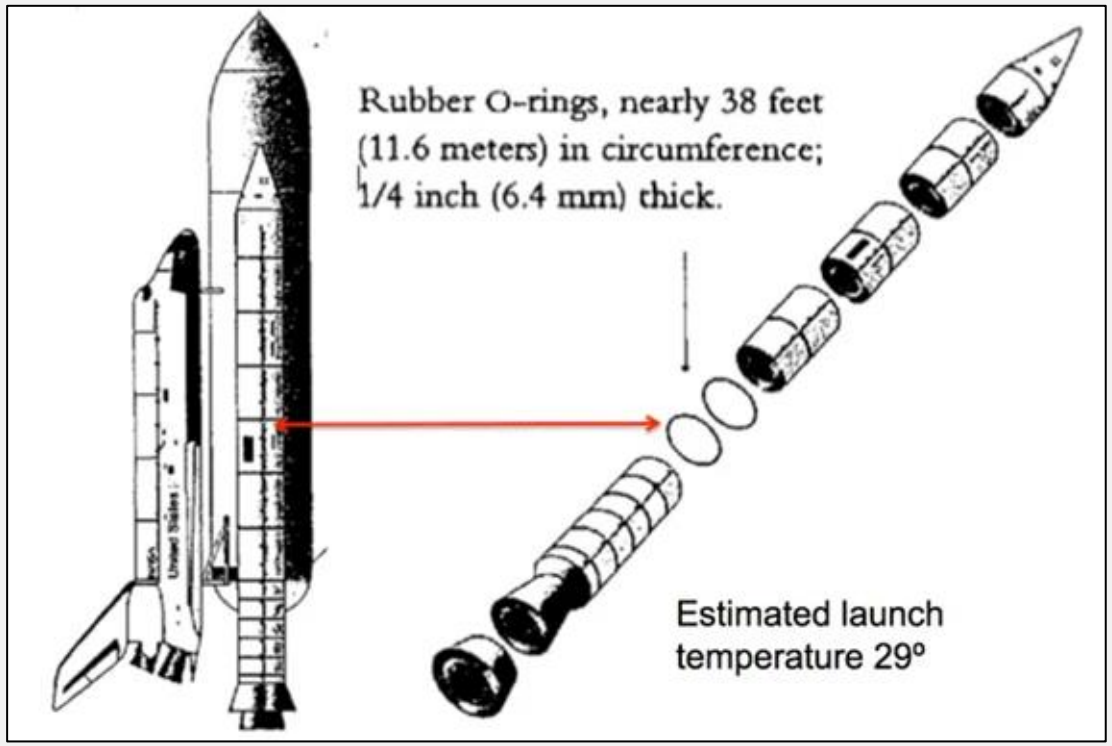
- The engineers had evidence of the risk of an O-ring failure.
- The data was presented in an incorrect and unconvincing way.

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Telling a data story correctly can save lives

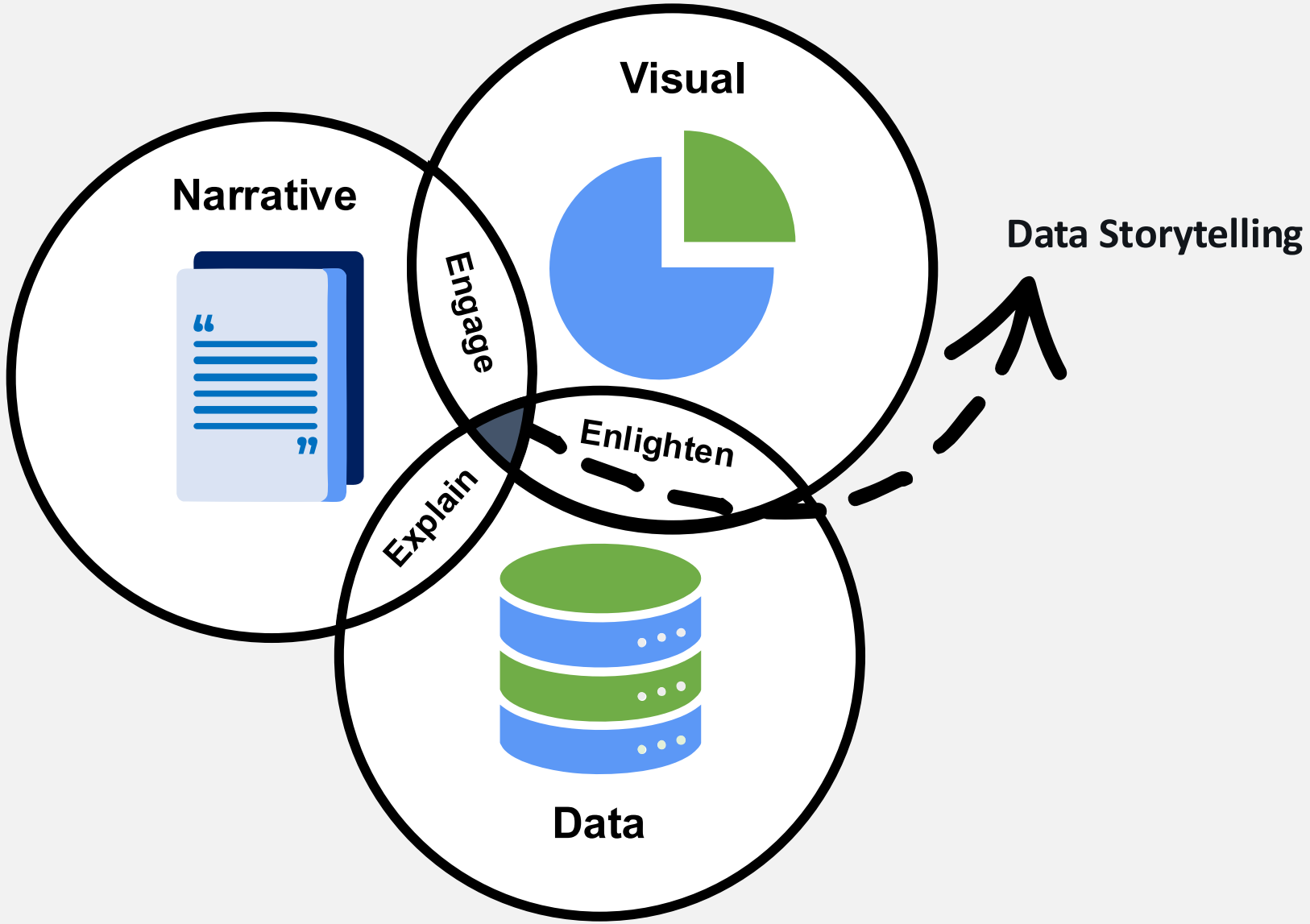
…or cost them



- The engineers had evidence of the risk of an O-ring failure.
- The data was presented in an incorrect and unconvincing way.
- The message didn't land and *astronauts lost their lives.*

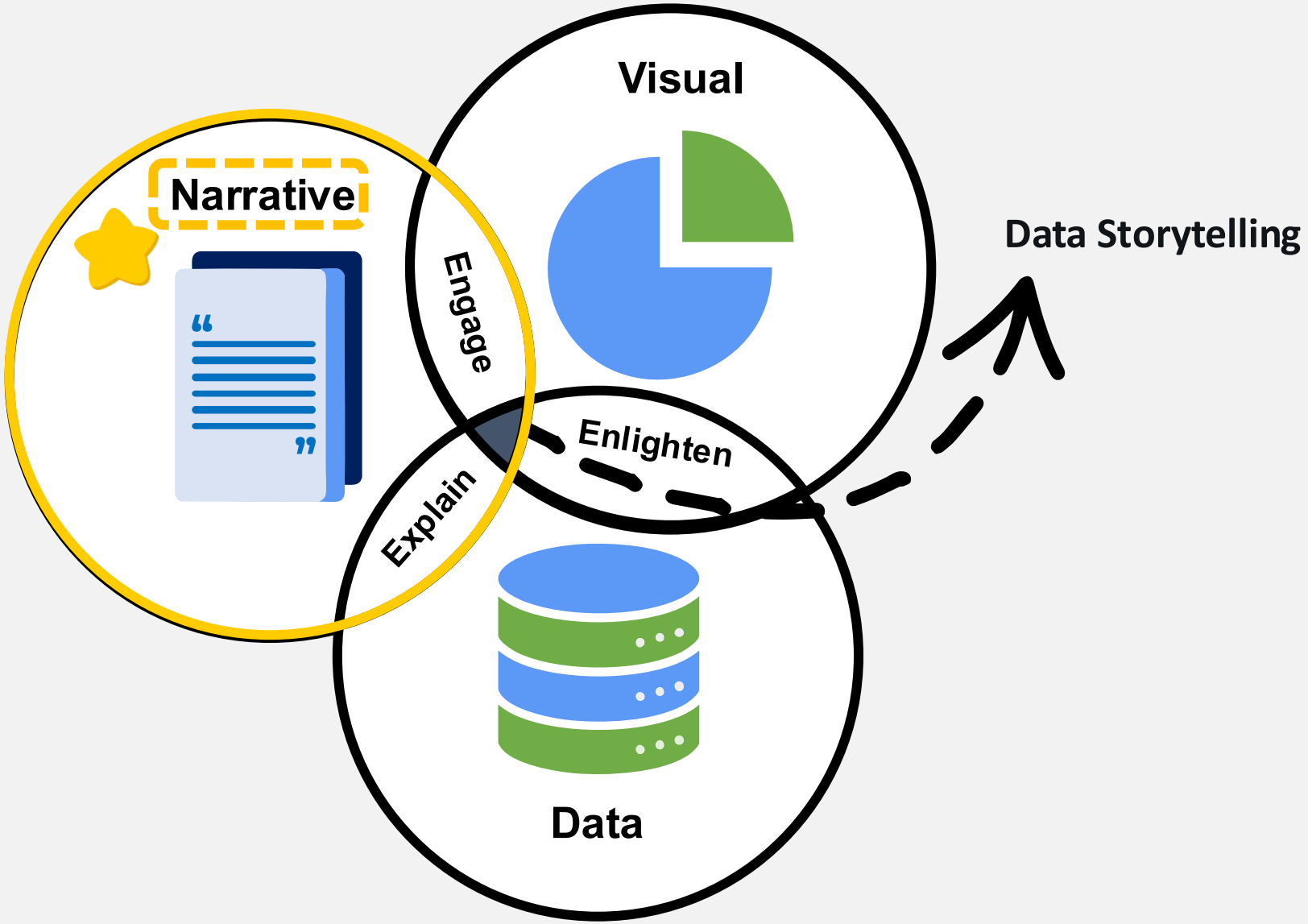
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Data Storytelling Types

Indirect Delivery

Asynchronous

- More detail (text)
- Less flexibility & control
- Exploration by reader
- Interpretation by reader

Infographics
Dashboards
Graphs

Direct Delivery

Synchronous

- Less detail (text)
- More flexibility & control
- Explanation of process
- Oriented to the audience
- Narrative

Presentations with Visuals that Support a Story

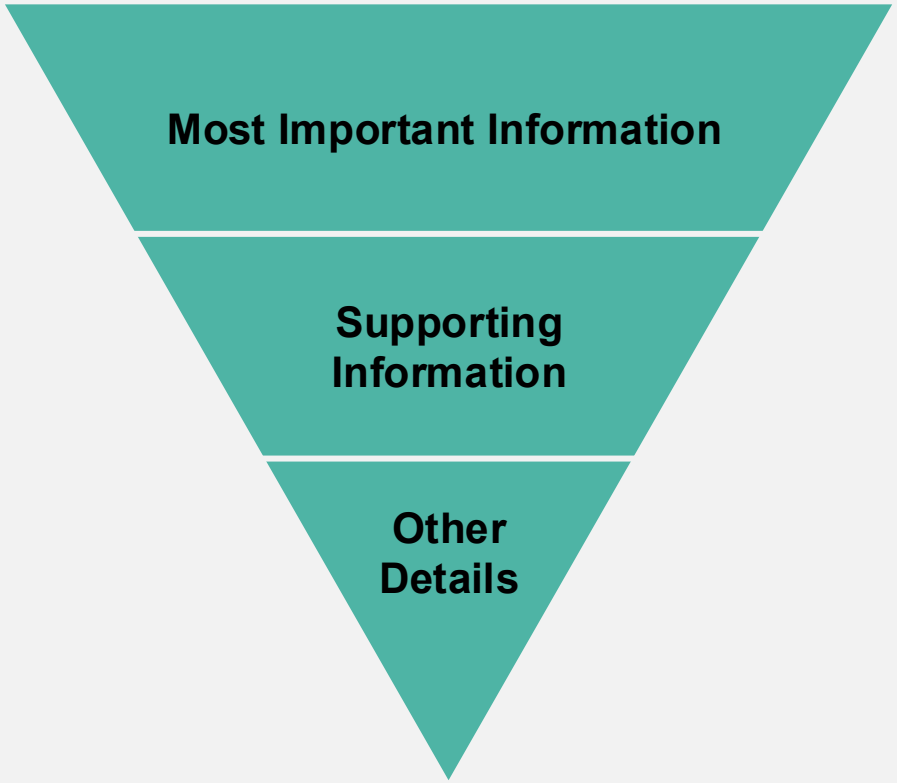
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Indirect Delivery vs Direct Delivery

Dashboard

Facts



- versus -

Data Story

Narrative

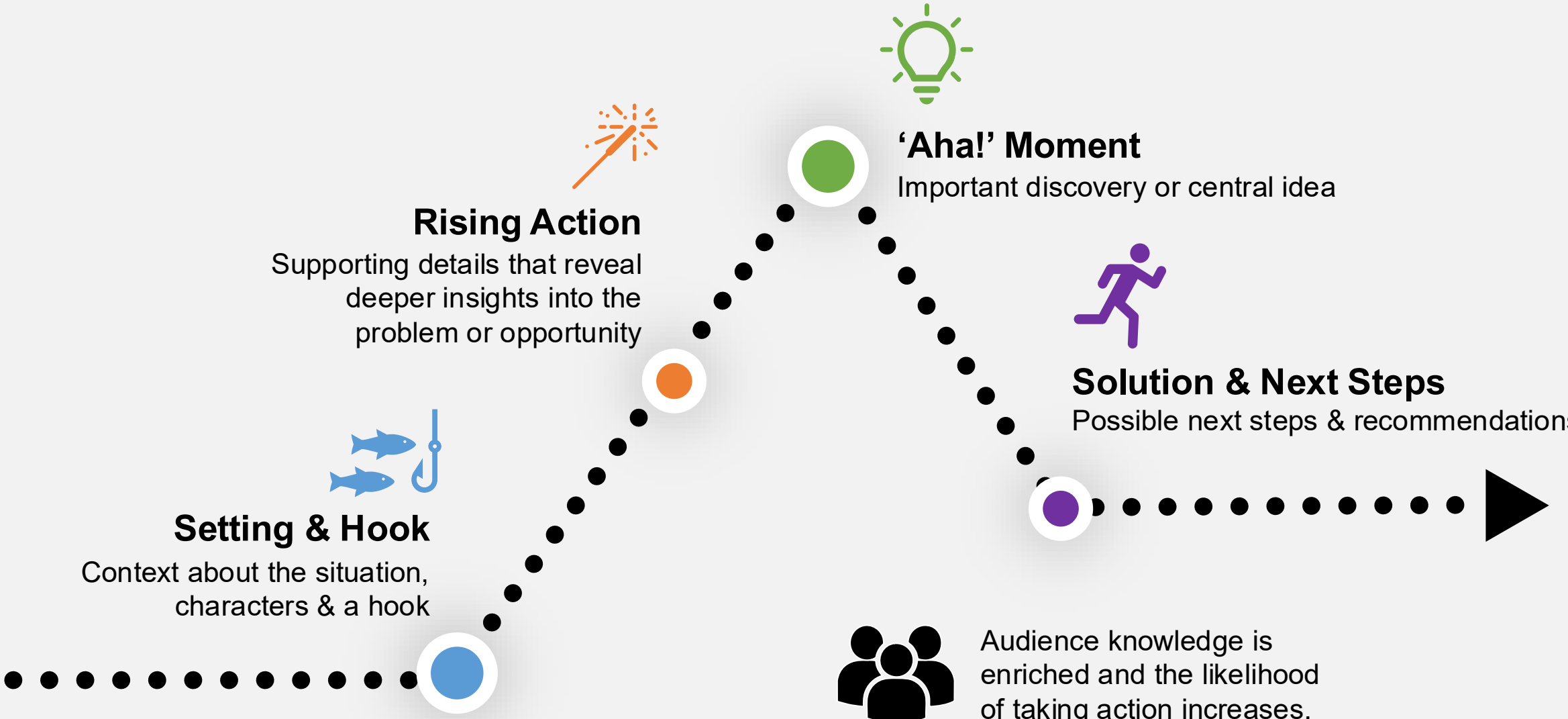


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Direct Delivery

Direct Delivery Flow



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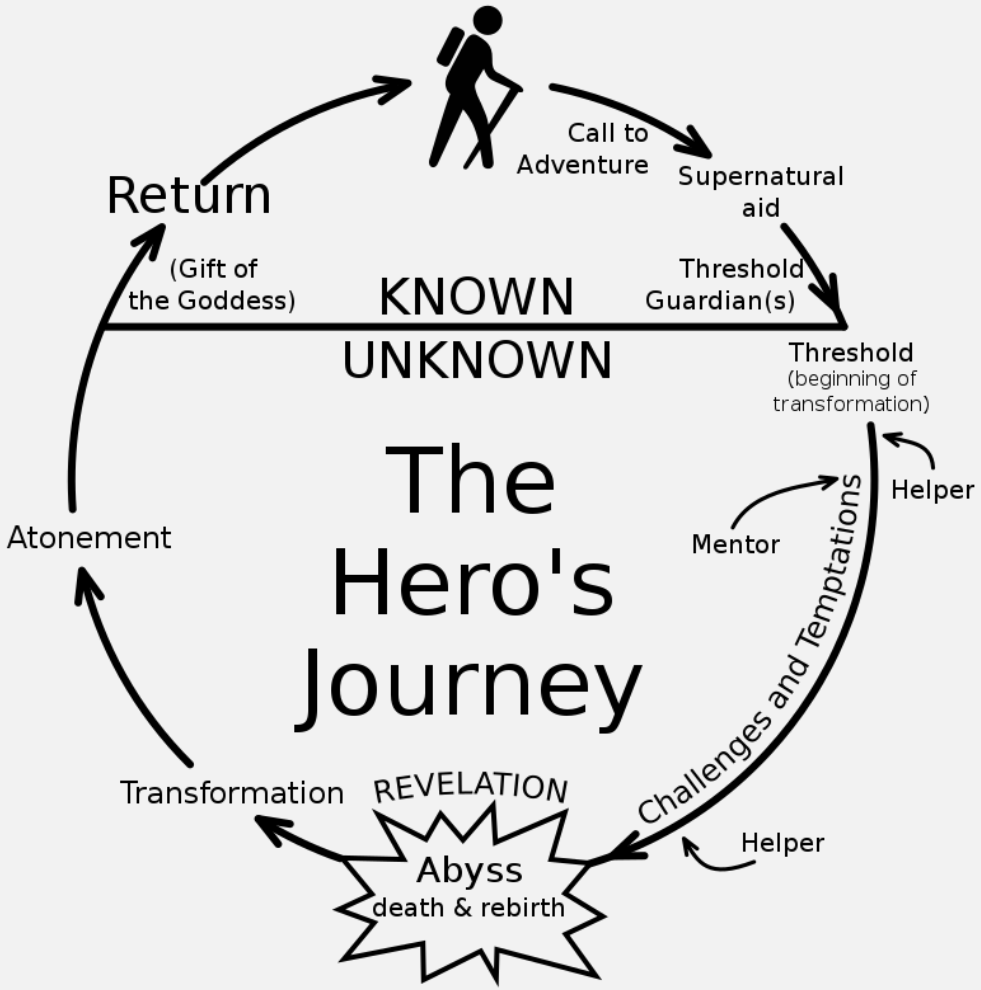
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TED-Ed
WHAT MAKES A
HERO?

Direct Delivery

Source: Phase5

Data Story



"

There is a **hero** who is faced with a **problem**.

The problem, if left unresolved, will result in negative consequences for either the hero or those the hero cares about.

In the hero's search for a solution, he or she is introduced to a **guide**.

The guide, often older and wiser, gives the hero a **plan** for how to overcome the problem.

Faced with the knowledge of what he or she must do, the hero is challenged to act via an **exhortation** (a calling or urging).

What follows is a **struggle** between the hero and the problem that ultimately leads to success or failure.

"

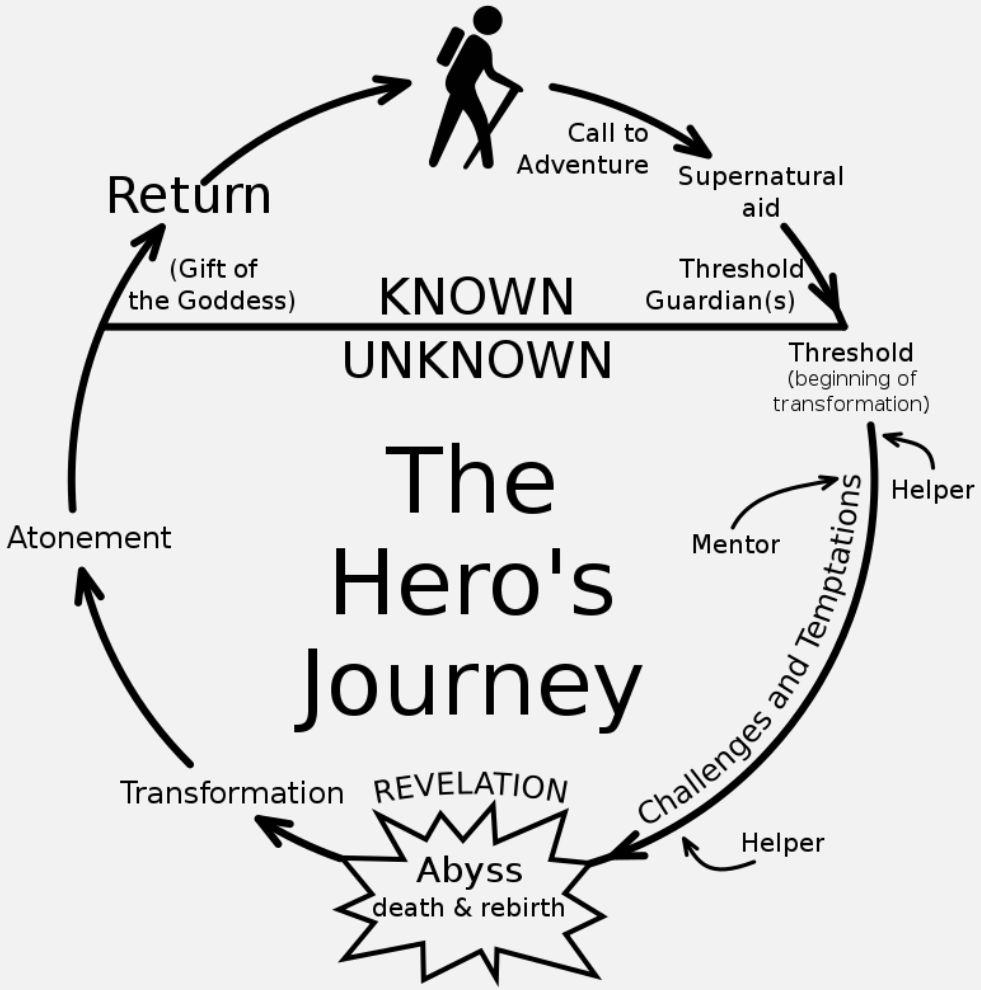
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Direct Delivery

Source: Phase5

Data Story



"

The hero: Your client

The problem: Your client's business problem that motivates the analysis

The guide: You

The plan: Your methodology and analysis

The exhortation: Your recommendations

Wise to avoid…

- Making yourself the hero
- Making the data the hero
- Making the analysis the hero

"

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Main Elements for Data Storytelling

1. **Characters:**
   Who are the stakeholders?
2. **Setting:**
   What is the context?
3. **Conflict:**
   What is the problem?
4. **Resolution:**
   What is the proposed solution?

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Mehrabian's 55-38-7 Communication Model

55%

of communication is **nonverbal**
(body language)

38%

of communication is **vocal**
(tone of voice)

7%

of communication is **words only**
(spoken word)



"Information is conveyed verbally, but in a face-to-face conversation, body language and facial expressions can have an incredible impact on how information is interpreted."

Source: THE UNIVERSITY OF TEXAS PERMIAN BASIN

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"For instance, research has shown that it takes conscious effort for a person to frown when looking at someone smiling – a smile is literally infectious."

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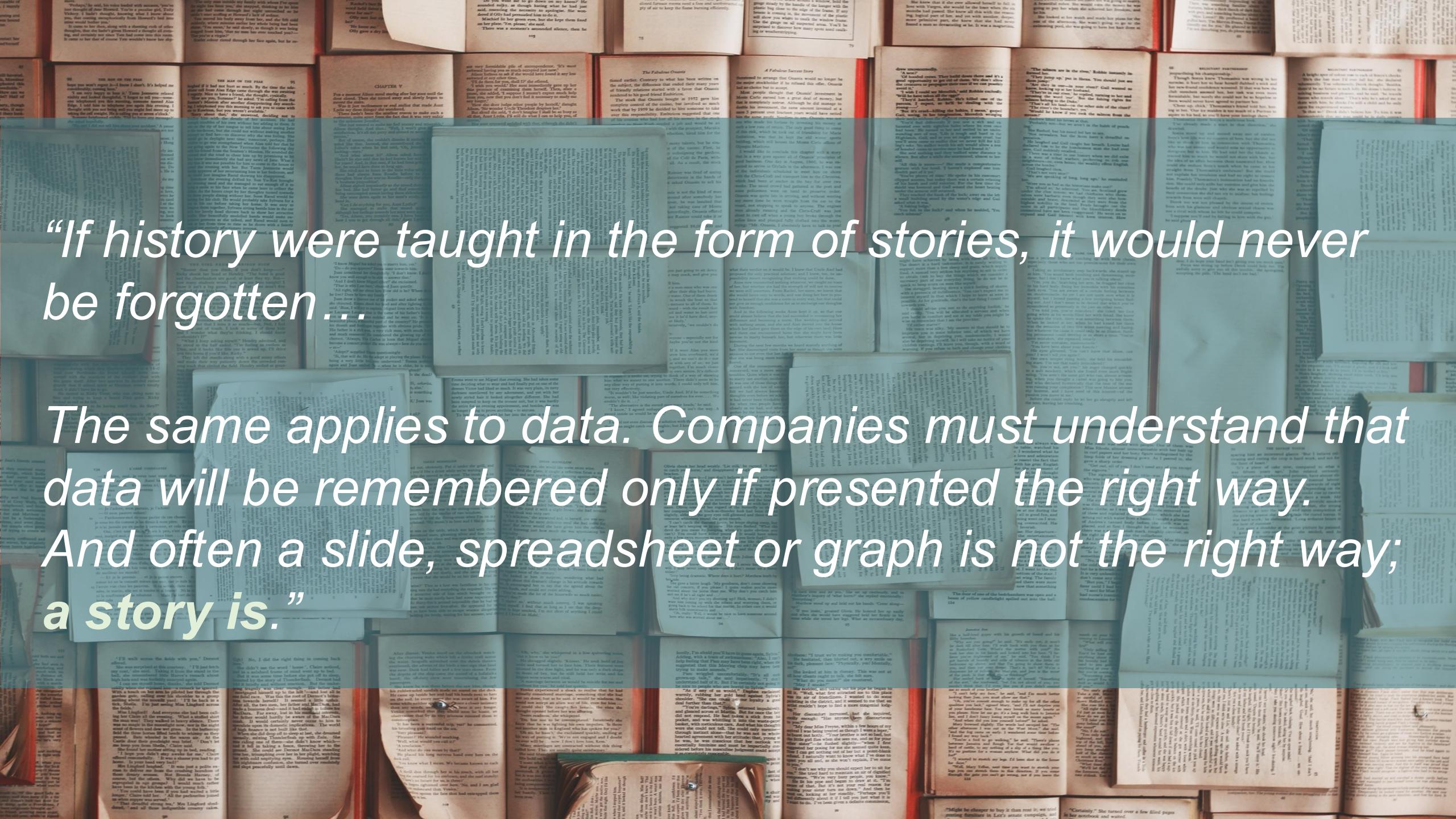
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Inspiration

Example narratives

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“If history were taught in the form of stories, it would never be forgotten…

The same applies to data. Companies must understand that data will be remembered only if presented the right way. And often a slide, spreadsheet or graph is not the right way; ***a story is****.”*

My Favorite Data Stories

“A Day in the Life of Americans”

- FlowingData
- http://flowingdata.com/2015/12/15/a-day-in-the-life-of-americans

“U.S. Gun Deaths”

- Periscopic
- http://guns.periscopic.com/?year=2018/

“What the World Eats”

- National Geographic
- https://www.nationalgeographic.com/what-the-world-eats/

”How many high school stars make it in the NBA?”

- The Pudding
- *https://pudding.cool/2019/03/hype/*

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A Day in the Life of Americans

Data used from the U.S. Bureau of Labor Statistics.

The American Time Use Survey asks thousands of people across the US what they did during a 24-hour period. The visualization simulates a single day for a sample of Americans representative of the population.. down to the minute!



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U.S. Gun Deaths

Near real-time, crowd-sourced data of U.S. deaths by shooting.

Surprising findings:

- Men & boys are the majority of victims – In 2010, 85% of those killed were men or boys.
- Young people are a significant portion of victims – In 2010, the majority of victims, 52%, were aged 30 or younger.

PERISCOPIC
DO GOOD WITH DATA

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What the World Eats

Daily diets vary widely around the world. Dive in to understand how diets vary from one country to the next.

Visuals are split by:

- Daily calories consumed per food type
- Daily grams consumed per food type
- Calories, grams and tons of meat consumed by meat type



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How Many High School Stars Make it in the NBA?

“Every year the top high school basketball recruits get hyped up. How often do they pan out?” This story visualizes how far high school stars from the past 20 years made it in their careers. Data is creatively represented with basketballs.



The Pudding

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More Inspiration I



"Why do dogs…?"

- Google Trends
- https://whydocatsanddogs.visualcinnamon.com/

"Examining Inequality"

- Bill & Melinda Gates Foundations
- https://www.gatesfoundation.org/goalkeepers/report/2019-report/#ExaminingInequality

"See Where Access to Clean Water is Getting Better – and Worse"

- National Geographic
- https://www.nationalgeographic.com/clean-water-access-around-the-world/#select/TOT/total

"Through the Eyes of a Survivor"

- Students in Harvard Data Visualization course
- https://hathix.github.io/through-the-eyes-of-a-survivor//

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More Inspiration II



“The Rhythm of Food"

- Google News Lab and Truth & Beauty
- https://rhythm-of-food.net/

“The Stories Behind a Line”

- A project by Federica Fragapane with Alex Piacentini
- http://www.storiesbehindaline.com/

“A Disappearing Planet”

- ProPublica
- https://projects.propublica.org/extinctions/

“Dear Data”

- Dear Data, The Project
- https://www.dear-data.com/theproject

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Feel free to stay in touch!

(I've reached the 30K connection limit on LinkedIn unfortunately, so until I can get that back down, you can follow me there!)

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